

Programs from Proofs III

Classical countable choice via products of selection functions

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Tutorial on programs from proofs

1. Ulrich Berger. Proofs as programs, program extraction in computable analysis, memoized functionals in higher types.
2. Monika Seisenberger. Examples of program extraction from intuitionistic and classical proofs that use choice: theory and Minlog implementation.
3. Martín Escardó. Giving a proof term for the classical axiom of countable choice via products of selections functions, and an illustration in Agda. Like Monika, I use the infinite pigeonhole principle as an example.
4. Paulo Oliva. Programs from classical proofs using Gödel's dialectica interpretation.

Plan for today

1. Selection functions.
2. Products of selection functions.
3. Double negation shift and classical countable choice.
4. Running a classical proof with choice in Agda.

Seemingly disparate constructions

1. **Topology.** Tychonoff Theorem:

X_i compact $\implies \prod_i X_i$ also compact.

2. **Higher-type computation.** Computational Tychonoff Theorem:

X_n exhaustively searchable $\implies \prod_n X_n$ also exhaustively searchable.

The point is that we get infinite exhaustively searchable sets.

3. **Game theory.**

Optimal plays of sequential games of unbounded length. Nash equilibria.

4. **Proof theory.** Double Negation Shift:

$\forall n \in \mathbb{N}(\neg\neg A(n)) \implies \neg\neg \forall n \in \mathbb{N}(A(n)).$

What do they have in common?

Implemented/realized by a certain countably infinite product of selection functions.

A higher-type functional that can be implemented in e.g. Haskell.

```
bigotimes :: [(x -> r) -> x] -> ([x] -> r) -> [x]
bigotimes (e : es) p = a : bigotimes es (p.(a:))
  where a = e(\x -> p(x : bigotimes es (p.(x:))))
```

Selection functions $(X \rightarrow R) \rightarrow X$

X set of things.

Goods in a store; possible moves of a game; proofs of a proposition; points of a space.

R set of values.

Prices; outcomes win, lose, draw; how much money you win; true or false; proofs again.

$X \xrightarrow{p} R$ value judgement.

How you value it; how much it costs you; pay-off of a move; propositional function.

$(X \rightarrow R) \xrightarrow{\varepsilon} X$ selects something according to some criterion.

The best, the cheapest, any, something odd.

Example 1

X set of goods.

R set of prices.

$X \xrightarrow{p} R$ table of prices.

$(X \rightarrow R) \xrightarrow{\varepsilon} X$ selects a cheapest good in a given table.

$(X \rightarrow R) \xrightarrow{\phi} R$ determines the lowest price in a given table.

Fundamental equation:

$$p(\varepsilon(p)) = \phi(p).$$

This says that the price of a cheapest good is the lowest in the table.

$$\begin{aligned}\phi &= \inf & \varepsilon &= \operatorname{arginf}, \\ p(\operatorname{arginf}(p)) &= \inf(p).\end{aligned}$$

Example 2

X set of individuals.

R set of booleans $\text{false} = 0 < 1 = \text{true}$.

$X \xrightarrow{p} R$ property.

$(X \rightarrow R) \xrightarrow{\varepsilon} X$ selects an individual with the highest truth value.

$(X \rightarrow R) \xrightarrow{\phi} R$ determines the highest value of a given property.

Fundamental equation:

$$p(\varepsilon(p)) = \phi(p)$$

$$\phi = \sup = \exists$$

$$\varepsilon = \text{argsup} = \text{arg-}\exists = \text{Hilbert's choice operator}$$

$$p(\varepsilon(p)) = \exists(p) \quad \text{Hilbert's definition of } \exists \text{ in his } \varepsilon\text{-calculus}$$

Maximum-Value Theorem

Let X be a compact non-empty topological space.

Any continuous function $p: X \rightarrow \mathbb{R}$ attains its maximum value.

This means that there is $a \in X$ such that

$$\sup p = p(a).$$

However, the proof is non-constructive when e.g. $X = [0, 1]$.

A maximizing argument a cannot be algorithmically calculated from p .

Of course, there is a **Minimum-Value Theorem** too.

Mean-Value Theorem

Any continuous function $p: [0, 1] \rightarrow \mathbb{R}$ attains its mean value.

There is $a \in [0, 1]$ such that

$$\int p = p(a).$$

Again this a cannot be found from p using an algorithm.

Universal-Value Theorem

Let X be a non-empty set and $2 = \{0, 1\}$ be the set of booleans.

Any $p: X \rightarrow 2$ attains its universal value.

There is $a \in X$ such that

$$\forall p = p(a).$$

This is again a classical statement if the set X is infinite.

This is usually formulated as the **Drinker Paradox**:

In any inhabited pub there is a person a s.t. if a drinks then everybody drinks.

We've also met the **Existential-Value Theorem**.

General situation

With ϕ among $\exists, \forall, \sup, \inf, \int, \dots$, we have that

$$\phi(p) = p(a)$$

for some a depending on p .

In favourable circumstances, a can be calculated as

$$a = \varepsilon(p),$$

so that

$$\phi(p) = p(\varepsilon(p))$$

Selection function

Definition.

A **selection function** for a (logical, arithmetical, . . .) quantifier

$$\phi: (X \rightarrow R) \rightarrow R$$

is a functional

$$\varepsilon: (X \rightarrow R) \rightarrow X$$

such that

$$\phi(p) = p(\varepsilon(p)).$$

Monad morphism

Every $\varepsilon: (X \rightarrow R) \rightarrow X$ is the selection function of some $\phi: (X \rightarrow R) \rightarrow R$.

Namely $\phi = \bar{\varepsilon}$ defined by

$$\phi(p) = p(\varepsilon(p)).$$

This construction defines a monad morphism $\theta: J \rightarrow K$:

$$\begin{array}{ccc} \overbrace{(X \rightarrow R) \rightarrow X}^{JX} & \xrightarrow{\theta} & \overbrace{(X \rightarrow R) \rightarrow R}^{KX} \\ \varepsilon & \longmapsto & \bar{\varepsilon} \end{array}$$

This is a morphism from the selection monad to the quantifier monad.

Oh, I mean to the continuation monad.

Units of the monads

$$\begin{aligned} X &\xrightarrow{\eta} KX \\ x &\longmapsto \lambda p.p(x). \end{aligned}$$

Quantifies over the singleton $\{x\} \subseteq X$.

$$\eta(x) = \exists_{\{x\}} = \forall_{\{x\}}.$$

$$\begin{aligned} X &\xrightarrow{\eta} JX \\ x &\longmapsto \lambda p.x. \end{aligned}$$

Produces a selection function for the singleton quantifier.

Functors of the monads

Let $f: X \rightarrow Y$.

$$\begin{aligned} KX &\xrightarrow{Kf} KY \\ \phi &\longmapsto \lambda p. \phi(\lambda x. p(f(x))). \end{aligned}$$

If ϕ quantifies over a set $S \subseteq X$, then $Kf(\phi)$ quantifies over the set $f(S) \subseteq Y$.

$$\begin{aligned} JX &\xrightarrow{Jf} JY \\ \varepsilon &\longmapsto \lambda p. f(\varepsilon(\lambda x. p(f(x)))). \end{aligned}$$

If ε is a selection function for ϕ , then $Jf(\varepsilon)$ is a selection function for $Kf(\phi)$.

Multiplications

They can be explained in intuitive terms, but this takes some time. (So I won't.)

$$\begin{aligned} \textcolor{blue}{K}KX &\xrightarrow{\mu} \textcolor{blue}{K}X \\ \Phi &\longmapsto \lambda p. \Phi(\lambda \phi. \phi(p)). \end{aligned}$$

$$\begin{aligned} \textcolor{blue}{J}JX &\xrightarrow{\mu} \textcolor{blue}{J}X \\ E &\longmapsto \lambda p. E(\lambda \varepsilon. p(\varepsilon(p)))(p). \end{aligned}$$

Use the selection function $\textcolor{blue}{E}$ to find a selection function ε such that $p(\varepsilon(p))$, and apply this resulting selection function to $\textcolor{blue}{p}$ to find an element of $\textcolor{blue}{X}$.

Monad algebras

$$KA \rightarrow A.$$

$$((A \rightarrow R) \rightarrow R) \rightarrow A.$$

Double-negation elimination.

Explains the Gödel–Gentzen translation of classical into intuitionistic logic.

$$JA \rightarrow A.$$

$$((A \rightarrow R) \rightarrow A) \rightarrow A.$$

Peirce's Law.

Get different proof translation of classical into intuitionistic logic.

Aside: we get a more conceptual explanation of **call/cc**

The **type** of the CPS translation of call/cc can be written as $JKX \rightarrow KX$.
(An instance of Peirce's Law, as first observed by Tim Griffin.)

Its **λ -term** can be reconstructed as follows:

1. KX is a K -algebra, with structure map $\mu: KKX \rightarrow KX$.
2. Because we have a morphism $J \xrightarrow{\theta} K$, every K -algebra is a J -algebra:

$$JA \xrightarrow{\theta_A} KA \xrightarrow{\alpha} A.$$

3. Call/cc is what results for $A = KX$ and $\alpha = \mu$:

$$JKX \xrightarrow{\theta_{KX}} K KX \xrightarrow{\mu} KX.$$

Strengths

$$\begin{aligned} X \times \textcolor{blue}{K}Y &\xrightarrow{t} \textcolor{blue}{K}(X \times Y) \\ (x, \phi) &\longmapsto \lambda p. \phi(\lambda y. p(x, y)). \end{aligned}$$

If ϕ quantifies over $S \subseteq Y$, then $t(x, \phi)$ quantifies over $\{x\} \times S \subseteq X \times Y$.

$$\begin{aligned} X \times \textcolor{blue}{J}Y &\xrightarrow{t} \textcolor{blue}{J}(X \times Y) \\ (x, \varepsilon) &\longmapsto \lambda p. (x, \varepsilon(\lambda y. p(x, y))). \end{aligned}$$

This produces a selection function for the above quantifier.

We have monoidal-monad structures

Because we have strong monads $T = J$ and $T = K$ on a ccc.

$$\begin{aligned} TX \times TY &\xrightarrow{\otimes} T(X \times Y) \\ (u, v) &\longmapsto (T(\lambda x. t_{X,Y}(x, v)))(u) \quad \longleftarrow \text{we want this one,} \\ (u, v) &\longmapsto (T(\lambda y. t_{Y,X}(u, x)))(v) \quad \longleftarrow \text{not this one.} \end{aligned}$$

The monads are not commutative.

The order in which you do things matters. (Illustrated below.)

Examples

$$\begin{aligned} \textcolor{blue}{K}X \times \textcolor{blue}{K}Y &\xrightarrow{\otimes} \textcolor{blue}{K}(X \times Y) \\ (\exists_A, \exists_B) &\longmapsto \exists_{A \times B}. \end{aligned}$$

$$\begin{aligned} \textcolor{blue}{K}X \times \textcolor{blue}{K}Y &\xrightarrow{\otimes} \textcolor{blue}{K}(X \times Y) \\ (\forall_A, \exists_B) &\longmapsto \lambda p. \forall x \in A. \exists y \in B. p(x, y). \end{aligned}$$

The other choice of $\otimes$ concatenates the quantifiers in reverse order.

Because we have a strong monad morphism:

$$\overline{\varepsilon \otimes \delta} = \bar{\varepsilon} \otimes \bar{\delta}.$$

In other words:

Theorem.

If

$\varepsilon \in JX$ is a selection function for the quantifier $\phi \in KX$,

$\delta \in JY$ is a selection function for the quantifier $\gamma \in KY$,

then

$\varepsilon \otimes \delta$ is a selection function for the quantifier $\phi \otimes \gamma$.

Binary product of quantifiers and selection functions

In every pub there are a man b and a woman c such that if b buys a drink for c then every man buys a drink for some woman.

Binary product of quantifiers and selection functions

In every pub there are a man b and a woman c such that if b buys a drink for c then every man buys a drink for some woman.

If X = set of men and Y = set of women, and if we define $\phi = \forall \otimes \exists$, i.e.

$$\phi(p) = (\forall x \in X \exists y \in Y p(x, y)),$$

then our claim amounts to

$$\phi(p) = p(a)$$

for a suitable pair $a = (b, c) \in X \times Y$,

This is calculated as $a = (\varepsilon \otimes \delta)(p)$ where $\bar{\varepsilon} = \forall_X$ and $\bar{\delta} = \exists_Y$, using the theorem.

The infinite strength of the selection monad

In certain categories of interest

There is a countable monoidal-monad structure

$$\bigotimes_n : \prod_n JX_n \rightarrow J \prod_n X_n$$

uniquely determined by the equation

$$\bigotimes_n \varepsilon_n = \varepsilon_o \otimes \bigotimes_n \varepsilon_{n+1}.$$

Turns out to be an instance of the **bar recursion** scheme.

The continuation monad **lacks** infinite strength

However, if a sequence of quantifiers ϕ_n have selection functions ε_n , then their product can be defined as

$$\bigotimes_n \phi_n = \overline{\bigotimes_n \varepsilon_n}$$

and uniquely satisfies

$$\bigotimes_n \phi_n = \phi_o \otimes \bigotimes_n \phi_{n+1}.$$

This is useful for various applications, including the double negation shift.

What does $\otimes$ do?

Many things!

1. **Designed** to implement a computational version of the countable Tychonoff Theorem.
2. It **turns out** to compute optimal plays of sequential games of unbounded length, and Nash equilibria.
3. It **happens** to realize the double-negation shift.
4. In the finite case, it gives Bekic's construction of fixed-point operators.
5. Among other things. See Paulo's talk tomorrow.

What does it do in general? Maybe (2) subsumes all cases.

Program extraction from classical proofs with choice

Start with intuitionistic choice

$$\forall x \in X (\exists y \in Y_x (A(x, y))) \implies \exists f \in \prod_x Y_x (\forall x \in X (A(x, fx))) .$$

Apply the T -translation, say for $T = K$ or $T = J$:

$$\forall x \in X (T \exists y \in Y_x (A^T(x, y))) \implies T \exists f \in \prod_x Y_x (\forall x \in X (A^T(x, fx))) .$$

Is that realizable?

 Does it have a proof term?

The J -shift

Think of $JA = ((A \rightarrow R) \rightarrow A)$ as a logical **modality**.

Theorem

The product functional $\bigotimes: \prod_n JX_n \rightarrow J(\prod_n X_n)$ realizes the **J -shift**

$$\forall n (J(A(n)) \rightarrow J(\forall n (A(n)))).$$

To guess the theorem, apply Curry–Howard.

To prove it, use bar induction and continuity.

Realizing the J -translation of countable choice

1. Start again with intuitionistic choice, but countable this time:

$$\forall n \in \mathbb{N} (\exists x \in X_n (A(n, x))) \implies \exists f \in \prod_n X_n (\forall n \in \mathbb{N} (A(n, fn))) .$$

2. Apply the functor J :

$$\underline{J(\forall n \in \mathbb{N} (\exists x \in X_n (A(n, x))))} \implies J\exists f \in \prod_n X_n (\forall n \in \mathbb{N} (A(n, fn))) .$$

3. Finally pre-compose with the instance of the J -shift

$$\forall n \in \mathbb{N} (J\exists x \in X_n (A(n, x))) \implies \underline{J(\forall n \in \mathbb{N} (\exists x \in X_n (A(n, x))))} .$$

Theorem. The J -translation of countable choice is realizable.

Realizing the K -translation of countable choice

It suffices to realize the K -shift.

However, we saw that K doesn't have a countable strength.

Reduce the problem to the J -shift.

We can go from J to K with the monad morphism.

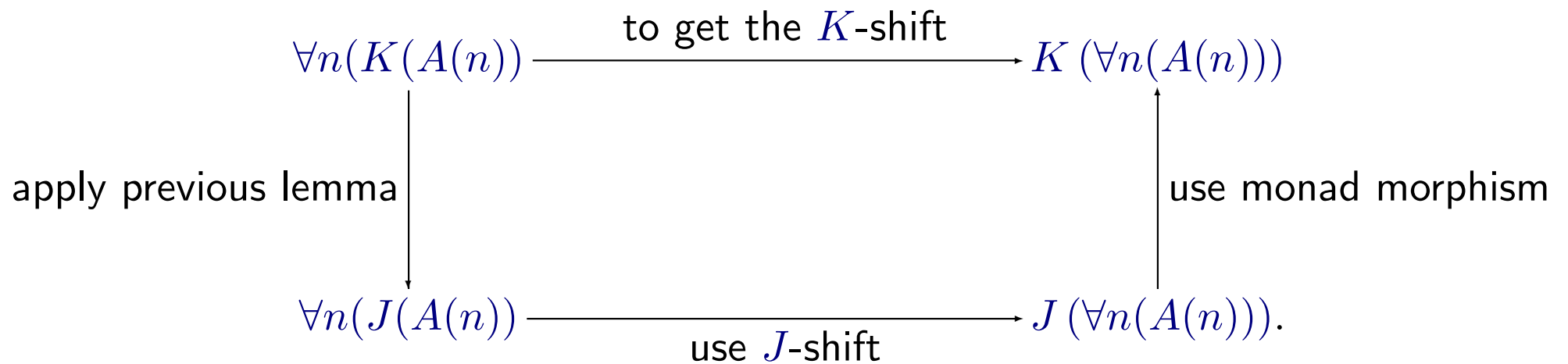
In general there is no way back.

But for formulas A in the image of the K -translation, we have $R \rightarrow A$.

Lemma. $(R \rightarrow A) \rightarrow KA \rightarrow JA$.

Proof. Transitivity of implication.

K-shift from *J*-shift



The *J*-shift is stronger than the *K*-shift, because it works for all formulas, not just the ones in the image of the translation.

Running a classical proof with choice in Agda

We program/prove the infinite pigeon hole in Agda.

If I run out of time, go to my web page, and click the link with this name.

I will include these slides there.

These slides also have an appendix with the human proof of the infinite pigeonhole principle.

Appendix

This should help you to understand the Agda modules

`http://www.cs.bham.ac.uk/~mhe/pigeon/html/InfinitePigeon.html`

`http://www.cs.bham.ac.uk/~mhe/pigeon/html/FinitePigeon.html`

`http://www.cs.bham.ac.uk/~mhe/pigeon/html/PigeonProgram.html`

`http://www.cs.bham.ac.uk/~mhe/pigeon/html/Examples.html`

even if you haven't seen Agda before.

Summary

1. From a brute-force classical proof of an infinitary statement, we discuss how to get a proof of a finitary statement that we can run.
2. And we run it, surprisingly, in Agda, an implementation of intuitionistic type theory.

There are two main steps:

1. Friedman's A-translation and Friedman's trick, for pure classical logic.
2. Use a proof term for classical countable choice.

We disable the termination checker in the module that defines the proof term of the *J*-shift.

Infinite Pigeonhole Principle: mathematical formulation

If **you** colour the integers with finitely many colours, then **I** can find infinitely many integers with the same colour.

This holds classically but fails constructively.

How do I identify the infinitely many integers that got the same colour?

We'll consider two colours without loss of generality, called **0** and **1**.

Then $2 = \{0, 1\}$ is the set of colours.

Infinite Pigeonhole Principle: symbolic formulation

$$\forall \alpha \in 2^{\mathbb{N}} \quad \exists b \in 2 \quad \exists g: \mathbb{N} \rightarrow \mathbb{N} \quad \forall i \in \mathbb{N} \quad (gi < g(i+1) \wedge \alpha(gi) = b).$$

1. The colouring you give to me is the infinite sequence α .
2. The colour I've found shared by infinitely many integers is b .
3. The subsequence of integers with that colour I've found is presented by g .

Infinite Pigeonhole Principle: deliberately brute-force proof

1. **Either** from some point on the given sequence α is constantly 0 **or not**.
2. **If it is**, we've found the desired colour and monochromatic subsequence.
3. **Otherwise:**
 - (a) For every n there is $i \geq n$ such that the colour of i is 1.
 - (b) By the **axiom of countable choice**, there is $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $fn \geq n$ for every n and the colour of position fn is 1.
 - (c) Then the desired monochromatic subsequence can be given by induction as

$$\begin{aligned}g(0) &= f(0) \\g(n+1) &= gn + 1 + f(gn + 1).\end{aligned}\qquad\text{Q.E.D.}$$

Infinite Pigeonhole Principle: symbolic version of the proof

1. Let $A = \exists n \forall i \geq n (\alpha_i = 0)$. By excluded middle, $A \vee \neg A$.
2. Assume A . Then the desired monochromatic subsequence can be taken as $g(i) = n + i$, with colour $b = 0$.
3. Assume $\neg A$. Then:
 - (a) $\forall n \exists i \geq n (\alpha_i = 1)$.
 - (b) By the **axiom of countable choice**, $\exists f: \mathbb{N} \rightarrow \mathbb{N} \forall n (f n \geq n \wedge \alpha_{f n} = 1)$.
 - (c) Then the desired monochromatic subsequence can be taken to be

$$\begin{aligned} g(0) &= f(0) \\ g(n+1) &= g n + 1 + f(g n + 1). \end{aligned} \qquad \text{Q.E.D.}$$

Infinite Pigeonhole Principle: avoiding “ $\geq$ ” for simplicity

1. Let $A = \exists n \forall i (\alpha_{n+i} = 0)$. By excluded middle, $A \vee \neg A$.
2. Assume A . Then the desired monochromatic subsequence can be taken as $g(i) = n + i$, with colour 0.
3. Assume $\neg A$. Then, **classically**,
 - (a) $\forall n \exists i (\alpha_{n+i} = 1)$.
 - (b) By the **axiom of countable choice**, $\exists f: \mathbb{N} \rightarrow \mathbb{N} \forall n (\alpha_{n+f(n)} = 1)$.
 - (c) Then the desired monochromatic subsequence can be taken to be

$$\begin{aligned} g(0) &= f(0) \\ g(n+1) &= gn + 1 + f(gn + 1). \end{aligned} \qquad \text{Q.E.D.}$$

Infinite Pigeonhole Principle: negative translation

General recipe:

Insert “ $\neg\neg$ ” in front of “ $\exists$ ”, “ $\forall$ ” and “ $=$ ”

When we can get away with it, we place fewer “ $\neg\neg$ ” than formally required.

In our example:

$$\forall \alpha \in 2^{\mathbb{N}} \quad \neg\neg \exists b \in 2 \quad \exists g: \mathbb{N} \rightarrow \mathbb{N} \quad \forall i \in \mathbb{N} \quad (gi < g(i+1) \wedge \neg\neg(\alpha(gi) = b)).$$

Facts about the negative translation

1. For any formula B in the image of the translation, $\neg\neg B \implies B$ has an **intuitionistic** proof.
2. For any formula A whatsoever, $\neg\neg(A \vee \neg A)$ has an **intuitionistic** proof.
3. In particular, for any formula A whatsoever, and any B in the image of the translation, $(A \rightarrow B) \wedge (\neg A \rightarrow B) \implies B$ has an intuitionistic proof.
4. From a classical proof of a formula, we automatically get an intuitionistic proof of its translation.

This fails if we have axioms that don't have as consequences their own translations, such as countable choice.

That's why we had to spend some effort constructing a proof term for the translation of countable choice.

Infinite Pigeonhole Principle: proof of negative translation

1. Let $A = \exists n \forall i \neg\neg(\alpha_{n+i} = 0)$. By **intuitionistic logic**, $\neg\neg(A \vee \neg A)$.
2. Assume A . Then the desired monochromatic subsequence can be taken as $g(i) = n + i$, with colour 0.
3. Assume $\neg A$. Then, by **intuitionistic logic**,
 - (a) $\forall n \neg\neg\exists i \neg\neg(\alpha_{n+i} = 1)$.
 - (b) By **classical countable choice**, $\neg\neg\exists f: \mathbb{N} \rightarrow \mathbb{N} \forall n \neg\neg(\alpha_{n+f_n} = 1)$.
 - (c) To conclude $\neg\neg$ -existence, we can take $b = 1$ and

$$g(0) = f(0), \quad g(n+1) = gn + 1 + f(gn + 1).$$

4. Therefore $\neg\neg\exists b \in 2 \quad \exists g: \mathbb{N} \rightarrow \mathbb{N} \quad \forall i \in \mathbb{N} \quad (gi < g(i+1) \wedge \neg\neg(\alpha(gi) = b))$
by **intuitionistic logic**. Q.E.D.

Infinite Pigeonhole Principle: Friedman's A-translation

General recipe: Choose an arbitrary proposition R , playing the role of false, and define $KA = ((A \rightarrow R) \rightarrow R)$ (generalized double negation).

Insert “ K ” in front of “ $\exists$ ”, “ $\forall$ ” and “ $=$ ”

When we can get away with it, we place fewer “ K ” than formally required.

In our example:

$$\forall \alpha \in 2^{\mathbb{N}} \quad K\exists b \in 2 \quad \exists g: \mathbb{N} \rightarrow \mathbb{N} \quad \forall i: \mathbb{N} \quad (gi < g(i+1) \wedge K(\alpha(gi) = b)).$$

Facts about the generalized double negation modality K

For any two formulas A, B , the following hold in intuitionistic logic:

1. $(A \rightarrow B) \rightarrow (KA \rightarrow KB)$ (strong functor)
2. $A \rightarrow KA$ (unit)
3. $KKA \rightarrow KA$ (multiplication)

It follows intuitionistically from the above that:

1. $(A \rightarrow KB) \rightarrow (KA \rightarrow KB)$ (Kleisli extension)
2. $(A \wedge KB) \rightarrow (KA \wedge KB)$ (strength)

Facts about Friedman's translation

1. For any formula B in the image of the translation, $R \implies B$ has an **intuitionistic** proof (ex falso quodlibet). The role of $\perp$ is played by R .
2. For any formula B in the image of the translation, $KB \implies B$ has an **intuitionistic** proof.

Translated formulas are algebras of the monad.

3. For any formula A whatsoever, $K(A \vee (A \rightarrow R))$ has an **intuitionistic** proof.
4. In particular, for any formula A whatsoever, and any B in the image of the translation, $(A \rightarrow B) \wedge ((A \rightarrow R) \rightarrow B) \implies B$ has an intuitionistic proof.
5. From a classical proof of a formula, we automatically get an intuitionistic proof of its translation.

Infinite Pigeonhole Principle: Friedman's translation

1. Let $A = \exists n \forall i K(\alpha_{n+i} = 0)$. By **intuitionistic logic**, $K(A \vee (A \rightarrow R))$.
2. Assume A . Then the desired monochromatic subsequence can be taken as $g(i) = n + i$, with colour 0.
3. Assume $(A \rightarrow R)$. Then, **intuitionistically**,
 - (a) $\forall n K \exists i K(\alpha_{n+i} = 1)$.
 - (b) By **Klassical countable choice**, $K \exists f: \mathbb{N} \rightarrow \mathbb{N} \forall n K(\alpha_{n+f_n} = 1)$.
 - (c) To conclude K -existence, we can take $b = 1$ and

$$g(0) = f(0), \quad g(n+1) = gn + 1 + f(gn + 1).$$

4. Therefore $K \exists b \in 2 \exists g: \mathbb{N} \rightarrow \mathbb{N} \forall i \in \mathbb{N} (gi < g(i+1) \wedge K(\alpha(gi) = b))$
by **intuitionistic logic**. Q.E.D.

Corollary: Finitary version

Every infinite sequence of two colours has finite monochromatic subsequences of arbitrary length.

Finite pigeonhole principle: symbolic version

$$\begin{aligned} \forall \alpha \in 2^{\mathbb{N}} \quad \forall m \in \mathbb{N} \quad \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} \quad & \forall i < m \quad (s_i < s_{i+1}) \\ & \wedge \quad \forall i \leq m \quad (\alpha_{s_i} = b). \end{aligned}$$

Using the Infinite Pigeonhole principle, we get

$$\begin{aligned} \forall \alpha \in 2^{\mathbb{N}} \quad \forall m \in \mathbb{N} \quad \textcolor{red}{K} \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} \quad & \forall i < m \quad (s_i < s_{i+1}) \\ & \wedge \quad \forall i \leq m \quad \textcolor{red}{K}(\alpha_{s_i} = b). \end{aligned}$$

Just take s to be the restriction of g . Now intuitionistic logic gives

$$\begin{aligned} \forall \alpha \in 2^{\mathbb{N}} \quad \forall m \in \mathbb{N} \quad \textcolor{red}{K} \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} \quad & \forall i < m \quad (s_i < s_{i+1}) \\ & \wedge \quad \forall i \leq m \quad (\alpha_{s_i} = b). \end{aligned}$$

Can shift $\textcolor{red}{K}$ out of finite quantifiers and arbitrary existential quantifiers.

Friedman's trick

Used to get rid of the last remaining K .

1. Suppose you have proved intuitionistically $K\exists x(Ax)$ for arbitrary R and that R doesn't occur in $A(x)$. (As we did.)
2. Then you get $\exists x(Ax)$ intuitionistically.
3. Indeed, $K\exists x(Ax)$ amounts to $((\exists x(Ax)) \rightarrow R) \rightarrow R$.
4. Now just take $R = \exists x(Ax)$ and we are done.

Finite Pigeonhole from Infinite Pigeonhole

In our example,

$$\forall \alpha \in 2^{\mathbb{N}} \quad \forall m \in \mathbb{N} \quad \textcolor{red}{K} \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} \quad \forall i < m \quad (s_i < s_{i+1}) \\ \wedge \quad \forall i \leq m \quad (\alpha_{s_i} = b),$$

take $\textcolor{red}{R}(\alpha, m) = \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} (\forall i < m \quad (s_i < s_{i+1})) \wedge \forall i \leq m \quad (\alpha_{s_i} = b)$
to get, intuitionistically,

$$\forall \alpha \in 2^{\mathbb{N}} \quad \forall m \in \mathbb{N} \quad \exists b \in 2 \quad \exists s \in \mathbb{N}^{m+1} \quad \forall i < m \quad (s_i < s_{i+1}) \\ \wedge \quad \forall i \leq m \quad (\alpha_{s_i} = b).$$

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